

"Danceability as an Approximate Measure of Groove Collective Synchrony"

33:33

all right that has concluded the live presentations section of the Symposium I'm now going to play the pre-recorded videos so one second to rearrange get it in position now we'll continue with the pre-recorded videos hopefully all of them concatenated into the next few hours okay this talk is by Leonardo MOA Rodriguez Groove as the musical glue for social bonding ladies and gentlemen hello welcome to today's talk on Groove as a musical glue for social bonding my name is Leonardo Mula Rodriguez I'm a PhD student at Center and this is a neuroscientific and crosscultural perspective now I'm a bit self-conscious by the fact that the only audio you'll be hearing today is my voice no music but do books sing No we're doing a deep dive into the scientific literature not musical theater maybe musical theater in the future of uh my active inference why my my name is Lear Müller Rodriguez I'm a PhD student at the Cambridge Institute for music therapy Research Center and my focus is on sleep music four dimensions and if you could how does this tie in with danceability where it relates to musical entrainment U more specifically slow beats I'm also a composer music producer and musician so how this groove Foster

social bonding there's a little outline
of today's presentation Groove and his
relationship to Spotify stability the
Neuroscience of it the social bonding
the crosscultural aspect flow States in
music what musical features Define
Groove an active inference application
of GrooVe and a I update to
Spotify uh suggestion so what is Groo in
essence it's the Apex of an inverted
u-shape relationship between rhythmic
complexity and grooviness so and
pleasure if you uh
grooviness is at a moderate level of um
rhythmic complexity which is um rhythmic
complexity is if you use the musical
term syncopation is the amount to which
the Rhythm deviates from the meter or
the beat
or alternatively you can think of um the
tempo the beat is a Tempo and it relates
to how certain you have your predictions
of that Rhythm so Groove is a
pleasurable sensation sensation of
wanting to move the body to music driven
by rhythmic patterns that balance
predictability and surprise here's
another study looking at the same thing
and we'll see into the Neuroscience in a
bit that proves this now spotify's
danceability is a measure they use and
they Define it as so we don't have the
the actual computation that they do it's
the suitability of a track for dancing
based on a combination a unknown
combination unless you work at Spotify
of tempo Rhythm stability beat strength
or pulse Clarity and overall regularity

and this data sets um someone put this on on reddit and they've looked at a million tracks and like okay you beat me in terms of data set um and so they've categorized all the musical genres several musical genres in terms of uh as a function of dance ability and based on the discussion and it makes sense um on Reddit it's more about rhythmic Simplicity seems to analyze because it's more more danceable music is techno deep house and then you have salsa music which is lower and people in the comments who are salsa Dan like what what is going on and especially metal heavy Rock or hard you know metal fans that were thinking like wait is comedy more dancable than my favorite wrer so they're a bit offended with that um at least sleep music makes sense that it's kind of low um yeah so it lacks this relationship between the complexity and predictability fun fact if you listen to all the million tracks it will take you 7 years and 9 months if you listen to it on 24/7 basis here's my data set I compared sleep soundtrack sleep playlists um that are either very popular on YouTube or that I've created my own self or playlist that I have millions of subscribers on Spotify to high energy music such as gy music that has this beast mode playlist has a mill 11 million uh subscribers so this regon uh music regon is a Latin popular dance music very popular 11 million and this has about 5 million and you see the dance ability is in general higher for

this higher energy music compared to sleep music and here I compared veillance and high energy music is also more positive apparently uh sleep music is more negative anyway let's dive into the Neuroscience of it and here the main thing I want you to focus on is the fact that b

perception is present in the brain and uh frequencies increase their strength at the onset of the pulse here there are different temp that were used or different speeds uh between pulses and as you see there's always an increase just after the uh pulse regardless of the tempo it happens all the time and it happens at the level of beta and Alpha brain waves and gamma brain waves these higher brain uh frequencies higher frequencies and um it activates these uh brain regions here and we'll go a bit more in detail but it seemed it's quite wide range and it turns out that we can call these the musical Rhythm brain Network because it makes use of the damic and rhythmic sorry the doer mintic reward system in the brain um for beat perception and positive effect of groovy music and this involves a supplementary motor area premotor area nearby the pmen as well as for pleasure and the urge to move to the music is the nucleus acaban and this connection to the medial orbital frontal cortex because we're talking about an auditory stimulus um auditory regions must be involved and uh the cereum also form fine motor movements turns out that musical Rhythm

predictions and surprises are measurable
in the brain in two forms
so the first one is about short-term
habituation to a rhythm via repetition
of a certain Rhythm and then the
deviation from that the surprising
change to that rhythm is seen here in
this paper where this this measure
called the mismatch negativity which
measures prediction errors in relation
to Rhythm deviations changes or Omission
that you don't have that note that used
to be in that
Rhythm and the other form of
surprises and predictions is by the
recruitment of long-term memory or uh we
can talk about it in terms of implicit
music theory syntactic knowledge where
here regardless of you're being told in
advance that there will be a
non-traditional chord sequence um that
will be played you will have this sort
of mismatch negativity which is more
specific because it recruits long-term
memory here is called the electric
response in the auditory nerve area and um
yes that's an interesting aspect um by
traditionally I mean traditional music
western music theory um for harmonies
here again we just looking a bit deeper
into this whereby beta waves we saw
earlier and gamma waves The High
Frequency waves we saw earlier in terms
of beat perception are responsible for
top down so the beliefs the predictions
better and the updating the surprises um
through the gamma high frequency
um activity in the brain

so another aspect of music is that it's usually done in groups such as musical performance um appreciation music therapy or learning about music and turns out that music in groups synchronizes brains and this is called interpersonal brain synchrony and in addition it recruits the same musical Rhythm Network that we just saw with a you know a bit more broadening of it it has the amygdala insula which is usually involved visceral um uh activity and more generally the prefrontal cortex the Motor areas uh you know could involve actually moving if you're performing and the basal ganglia in general which incorporates all the dopaminergic reward system which includes uh and dopaminergic system Etc putamen in Cate we saw that

earlier now music the real function of Music beyond the auditory cheesecake that Stephen Pinker says

it's not just pleasure not just reward systems it has to do with social bonding and building trust music is a participatory um activity and dancing actually is a way to

participate without any musical virtuosity or even familiarity to the music this is why I call it embodied Democratic music participation open to all and fosters joy and social bonding and music strengthens this interact uh interactive rituals you can talk about it in terms of ritual and it's a social tool for enhancing uh groups so as as you can see here you

start to predict you the Beats you see
repetitive motions this induces Groove
induces dancing to it and this enhances
social bonding within this sort of you
can frame it within an agent environment
system

more of a uh seeing others uh dance and
move encourages dancing so this
reinforces the social bonding aspect and
this was exemplified here where eyes
open uh with low Tempo High Tempo High
roof actually increases

the collective movement um in a group of
people listening to music and this is
what they were measuring here with this
cross

wavelet

and uh dance promotes also here
interpersonal movement interaction
promotes self other integration it
enhances social cohesion through
synchronized movements and more
specifically

moderate uh syncopation and low as well
for AI there an implication for parkon
disease low and moderate syncopation
which is groovy

music um does so as well it

encourages self other

inclusion Which social B and now we're
talking about music The Groove and is it
applicable to a crosscultural uh

definition of Music well first there has
been this is with Japanese music

actually there has been an optimal

Groove Tempo range identified um which
seems to be a u inverted u-shape
relationship non linear relationship

with Tempo is this seems to be from 107

to

126 B BPM beats per minute uh Tempo and

The evolutionary interpretation for this

is the fact that it's near 120 BPM which

is the optimal tempo for walking and

dancing so this is where Groove might

come into sensory motor synchronization

is most optimal in that around 120

BPM um and there are also universals

crosscultural universals of Music which

has been identified using statistics

um and the main aspects that are

crosscultural is Beats dancing making

music in group percussion repetitive

melodic sequences these are musical

features that are Universal just going a

bit deeper into this they used a music

for all around these spots it's quite a

geographically varied data set although

only 34 recordings you know there's

millions of music but I think it's a

nice General generalization we make

based on the Black Box the black box

here are musical features that are at

least

50% uh shared in globally and the most

uh which are the most you know the

whiteness as is

shared um the most share of the beat

subdivisions by binary tary subdivisions

or scales that have up to seven notes so

this is interesting and this was

controlled phenetically meaning that

this ensure that the these musical

patterns and associations are not solely

due to Shared cultural or evolutionary

histories another aspect that is

crosscultural are musical bodily
Sensations and emotions what I want you
to focus on here is this correlation
they asked um Asian participants 5 500
people and Western participants 500
people to associate emotional labels to
musical features and there's a very
strong correlation
094 uh between these so there's a sort
of crosscultural aspect here obviously
it's not all cultures maybe African
Latin and uh Middle Eastern cultures
do feel differently but here's just a
Latin um influence that's indeed
instrumental sinking in syn pated music
which is uh groovy music which here was
Samba um encourages the urge to dance
so Groove at least is present in Latin
American music now musical flow is not
restricted to groove what is flow flow
is a state of absorption total
absorption in an activity in other words
the feeling of losing yourself in the
moment uh flow state is usually
characterized as a state of emotional
resilience against traa
and here emotional expressivity is one
of those not Groove uh uh musical
features that induces musical flow and
here the two uh main things I want you
to focus on is the blue um histograms
which are state of flow here were
performers piano performers and this is
a musical feature and so there was
higher flow during swelling dynamics of
crescendos that
which is volume um uh
fluctuations as well as um

repetition of um

Melodies and not only just any Melodies
but stepwise Melodies that are not too
far apart in terms of distance on the
keyboard uh or in terms of notes so the
closer they are blue is again flate the
closer they are the The Melodies the
more flow and juicing

they

provoke um another musical element that
provokes flow that is not Groove is the
narrative Arc storytelling emotional
Journeys in music and this interesting
paper by colleague Annie hesit
identified stages emotional stages of
the heroes Journey which is a a
narrative Arc found in myths films
popular films yeah in stories narratives
in general they've identified emotional
stages of the hero's journey in patients
and clients specifically in this case
was eating disorders but applying to
just more

generally undergoing music therapy so
most they were look they were doing
guided imagery with music which is a
type of music therapy

and depending what they're saying the
clients these themes were identifiable
within this uh the heroes Journey
narrative Arc so that's very interesting
um and this is very important for music
therapists because this uh informs
further the scanning aspect that they
usually do in

gim which is that they try to identify
using the music where is the emotional
stage that the client and the therapist

should start their um work emotional
work and so this is actually a emotional
Journeys are actually a concept that is
applied by DJs they usually talk about
energy arcs for their DJ sets um and
it's also very prevalent in film
soundtracks um you know film
soundtracks are using so music is being
used to guide the emotional journey of
the
audience
now flow as neural
entrainment I think it's neural
entrainment is evident for beat
perception we've seen it several times
here's more evidence like accurate beat
perception increases neural entrainment
here is just saying that beat
sensitivity uh there's a sort of tempor
range where this sensory motor
synchronization decreases uh with very
slow Tempe um but here again be
perception this the beat is being um
being identified in a neural activity
and even if you imagine certain rhythms
these frequencies appear also in the
brain activity so neuron training with
beat perception is clear but how about
for emotional expressivity and um
emotional Journeys well this
study um reconstructed the music that
was listened to by the uh participants
from their neural patterns it was a
combination of EG and fmri a technique
that made use of EG and fmri and and
deep neural networks to reconstruct the
music based on the neural
activity and you can see that not only

very impressively resembles quite a lot in terms of spectrogram information but obviously there's not just Rhythm activity or yeah Rhythm information that's being displayed there's more there's frequency activity so this takes us to this study that demonstrated that spectral flux is actually the most um accurate measure for neural entrainment to music and you can see this with this red line is the highest correlation um and with neural entrainment and in addition they've uh demonstrated that it is a metric that encompasses amplitude Envelope as well as beats they they have mutual information here and spectral flux or spectral novelty is called is a measure of Novel energy increases in the frequency spectrum um so this an interesting aspect now nevertheless as you can see other strong neural synchronization musical features um in addition to spectral flux are Tempo range so uh roughly as you can see here with the orange uh line 60 to 120 BPM uh High familiarity with the music and post CL way to perceive the Beats so let's keep it simple what are the groove's musical features two things moderate rhythmic complexity within an optimal range of around the walking tempo of humans um this these two Encompass most of these characteristics syncopation Tempo micro timing event density this has to

do with and even beat sense has to do with a more complex complex soap of attack is a variant of beat saliency a low frequency band is the base enhances um dancing uh volume and then this is the core progression and that now keep in mind that sensory motor synchronization is not limited to this uh Groove Tempo range the sensory motor uh synchronization Tempo range is shown to be between 60 to 180 BPM below Above This the sensory motor synchronization which is usually finger tapping or dancing to the beat decreases um which is probably linked to biomechanical reasons um however probably training increases your capacity to do so uh to synchronize however the beat perception still can occur Beyond those uh tempo this Tempo range uh I recommend watching this YouTube video by Adam NE which uh goes into this uh the extremes of beat perception let's apply active inference into Groove but first before we do that we need to understand that the brain is constantly generating predictions about how the world Works in order to minimize the uncertainty about how the world Works two the brain is a lay thing and it balances top down and bottom up influences where higher layers send beliefs about the world down to lower sensory layers which the sensory layers as we can see here um actually send back up sensory stimuli to adjust the uh beliefs through the mechanism of minimization of

prediction errors or minimization of
free

energy now talk about groove groove
balances confid

prediction uh confident predictions with
surprise about the Rhythm uh just
reminding you it is an optimal playing
with belief updating about musical
rhythms now listening to groove is
accessible to groups of people and
therefore can become a shared stimulus
for motor and emotional synchrony and
therefore under the active inference uh
framework

um generates generalized
synchrony uh which
means that people have shared World
models the listeners have shared World
models

and this is depicted

here um uh with this quote it say
neither a model of my behavior or your
behavior but a model of our Behavior
quote by

friston

um and generalized synchrony actually
requires flexibility in terms of your
higher uh

beliefs um and because you're trying to
model uh flexibly your Collective
identity and Collective behavior and
because of this and the social bonding
aspect I place musical groove on the
spectrum of the Rebus hypoth hypothesis
which is a relaxed belief under
psychedelics meaning that it's not as
the the flattening of the free energy
landscape is not as Extreme as

psychedelics are supposed to do but musical Groove can achieve this through the the social bonding so dancing to the same music not only increases trust and social bonding but it also allows for individual creativity compare an expert dancer and a novel dancer um a novice dancer and an expert won't be having motor mimicry but they will be sharing emotional s and this uh provides the mechanism for trust and bonding while considering uh creativity of individuals and you can make use of a Federated belief multi-person system um model Groove as the shared marginal likelihood that people nonverbally communicate um so because idiosyncratic dance moves have been shown to even increase more uh the social bonding and Trust I encourage you next time to take the risk and bust on some moves on the on the dance for all right so what are what I want you to consider right now is the current models of the context that encourages dancing yes they say musical properties they have temporal predictions that you can do induces pleasure induces arousal and together with regularity Chris urged to move to the music and um this will generate motor planning or policy generation under the active inference uh framework and and this desire to move and in training um body movements creat social connection but I like to add a bit more to this in

terms of contextual factors based on the literature review we're presenting today this is a hypothetical contextual Factor um framework for dance ability um basically Groove is the mechanism for uh dancing but there are several contextual layers that people individuals might need to cover um before starting to dance for instance I have two example individuals a white individual here this white uh Arrow who requires um seeing other people seeing their friends to before starting to move whereas this red individual just requires to feel a change in emotions to start wanting to move and these layers uh have a nonlinear relationship with the group meaning that uh your entry level contextual entry level to start dancing um might be seeing a friend starting to dance and you might not need familiarity with the music to start dancing to start connecting to the groove uh whereas DJs sometimes make use of familiar music in their DJ sets to encourage people to start dancing they say getting those first three people to dance is always the most difficult part and uh once you have them the crowd joins and you have a crowd dancing so this was modeled with a reduction of entropy to do it relatively simply uh whereby depending on the distance to the groove there's a higher um entropy level that has a nonlinear relationship uh with grooving and K has is based on personality and the relationship between

each

layer there's another way of

conceptualizing this as a the gravity of

Groove the layers uh are done like this

so implications for Spotify's

danceability we saw that is more likely

has to do with rhythmic complexity

comedy is more danceable than

uh Blues apparently or or heavy metal

headbanging is not as you know dancing

as as

laughing veence as well the most joyful

music seems to be as Sala and the less

joyful music is sleep maybe Spotify is

inspired by uh the rapper Nas who has a

lyric saying sleep is a cousin of

death so let's go beyond this and uh and

uh on Rhythm and Tempo and incorporate

what we've been seeing today into dance

ability so if you want to generate the

formula that goes beyond the ability

unknown equation which is most likely

rhythmic Simplicity we

convert uh we Sorry first uh I want to

point you to this study which asked

musicians to perform groovy music uh

with simple and complex M Melodies and

they show that there's about 30%

occurrence of fast notes uh specifically

16th notes um when people perform groovy

music and this was also confirmed in

terms of listeners saying yeah that's

groovy um and so here I've just uh used

the optimal Groove range and transformed

it into seconds so you can have an idea

and ah too quick for Spotify they can't

no I'm just kidding uh you get to see

the the the equation um so what I've

done is I made use of uh spectral flux
which is this spectral novelty makes use
of for transform and then logarithmic
compression to enhance uh weakened
spectral components and then you compute
the difference between time frames in
terms of positive energy
increases and so the measures positive
increases in Energy Event density is
another measure that's been used in
these studies and it's a number of
events onsets per a Time window so
here's a Groove uh formula I suggest
which is open to to your feedback um
which is 30% of energy
onsets uh that is in the 16th of a note
um time
window um divided by a
measure that's dependent on the tempo
and a measure is four times a beat
because most music is on 4x4 meter music
so I want to give you this quote dance
is an energetic mode of participation
for a large population regardless of age
familiarity with music or instrumental
singing
virtuosity to summarize we found that
Groove is a musical mechanism for social
bonding through movement and therefore
through movement is danceability right
and we can make use of the active
inference to model grp as a musical
mechan mechanism for generalized
synchrony even uh as Federated beliefs
to um model individual creativity
further Works um would encourage the
usage of uh um other social bonding
metrics in music such as group

singing uh chanting is seen around
rituals around the world and this has
implications for group music therapy
life performances and team building
music recommend assist sys Etc so please
do comp contact me I'm open to
collaborate for this uh Groove um
projects and discuss any of your ideas
uh we could discuss about musical uh uh
you know exchange music if you want and
I just want to show here my YouTube
channel which explores thanks to the
YouTube channel I've actually learned
how active inference works you know they
say teaching is the best way of
memorizing and in here for instance I
made explored Chris Field's um work in
the active inference with the
holographic
boundary um for social interactions
which is which could also be a a further
uh direction of of trying to to
conceptualize musical Groove in that
manner but shared marginal likelihood is
probably very effective as well uh thank
you so much please do ask me for the
references um I will send them they have
been sent or yeah I should be on the
chat by way asking answering some
questions thank you very much for your
time see you very soon byebye thanks